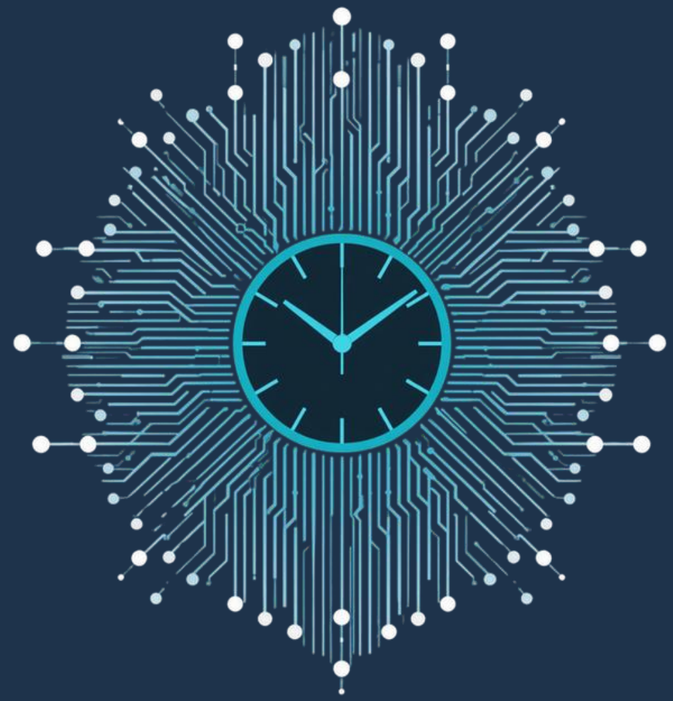
EMBEDDED SYSTEMS • FUNDAMENTALS

RTC (Real Time Clock)

Fundamentals and Applications in Embedded Systems



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What is a Real Time Clock?

A dedicated hardware clock that continuously tracks real-world time and calendar data—even when the main system is inactive.



01

Accurate time + date

Maintains seconds, minutes, hours and calendar information.

02

Processor independent

Continues running without firmware or CPU intervention.

03

Backup powered

Typically supported by a coin cell or supercapacitor.

Why RTC matters

Time becomes a dependable system resource—not a software assumption.

01

Keeps time when main power is off

02

Time-stamps events and system logs

03

Drives schedules, alarms and timers

04

Fits low-power and battery devices

How RTC works

A stable low-frequency reference is divided, counted and exposed to the host processor.

Crystal oscillator

32.768 kHz

Frequency divider

÷ 2^{15} → 1 Hz

Counter registers

sec • min • hr • date

I²C / SPI

host interface

Backup power

Battery / supercapacitor

The backup domain keeps the oscillator and counters alive while the main rail is absent.

Common RTC ICs

Choose for accuracy, power budget and integration—not familiarity alone.

LOW COST

DS1307

I²C • external crystal
Basic calendar RTC

HIGH ACCURACY

DS3231

TCXO • ±2 ppm class
Temperature compensated

LOW POWER

PCF8563

I²C • low standby current
Portable-device favorite

Applications of RTC

Wherever events need context, RTC provides the when.



01

Embedded systems

Industrial controllers and appliances

02

IoT devices

Smart sensors and wearables

03

Computers + servers

System clock and BIOS

04

Consumer electronics

Cameras and DVRs

Advantages & limitations

ADVANTAGES

01 CPU-independent timekeeping

02 Low power with backup supply

03 Simple I²C / SPI interface

04 Long operational life

LIMITATIONS

01 Crystal quality affects accuracy

02 Backup battery ages over time

03 Clock / calendar scope only

04 Drift without compensation

Reliable time. Independent power.

RTC is a critical component for reliable timekeeping in embedded systems and beyond.



Questions?